who only receive core data would be better able to observe if a smaller order steps ahead of their order at a better price and may be able to adjust their limit order in response.

One commenter stated that protecting smaller round lot quotes would negatively impact the trading of institutional investors because market participants would post smaller displayed quotes and institutional investors with larger orders would have to slice their trading activity into smaller increments to avoid signaling their trading interest.[1872] The Commission does not believe that protecting the smaller round lot size in higher priced stocks will negatively impact the trading of institutional investors. It is already common practice for institutional investors' parent orders to be sliced into smaller child orders.[1873] Additionally, because the round lot tiers are based on a notional value, $10,000, which is larger than the average trade size, $8,068 (excluding auctions),[1874] the Commission does not believe that market participants are likely to significantly reduce the size of their displayed limit orders and institutional investors' orders will not have to be sliced into smaller sizes than they already are. Additionally, the Commission believes that protecting the smaller round lot sizes in higher priced stocks could benefit smaller odd-lot-sized child limit orders that institutional investors submit. Because more of these orders would now be observable in core data, they may be more likely to execute against the marketable orders of market participants who rely on SIP data and were not previously able to observe these orders, as described above in this section.

1872 See T. Rowe Price Letter at 3.

1873 See supra Section V.B.3(e).

1874 See supra note 1862 and accompanying text.

Commenters stated that the Proposing Release did not consider the effects the smaller round lot size could have on the options market, where the standard options contract size is 100 shares.[1875] The Commission does not believe the new round lot tier sizes will have a significant impact on the options market for a number of reasons. First, the new round lot size will not change the size of the options contract. Second, most NMS stocks will still have a round lot size of 100 shares under the new round lot tier definitions.[1876] Third, even for stocks that are in a higher priced round lot tier, the smaller round lot may not have a significant impact on quoting in the options market because the round lot definition will not change market maker quoting obligations in the options market. Fourth, because there is already a significant presence of odd-lot quotes better than the NBBO in higher priced stocks,[1877] the best prices in these stocks are already frequently smaller than 100 shares. Therefore, the change in the round lot size may not have a significant impact on arbitrage opportunities between the options and equity markets for stocks in the higher priced round lot tiers. Fifth, the options markets already have standard options contracts on stocks with a round lot size less than 100 shares, so there are already conventions for dealing with options in which the round lot size in the equity market is not 100 shares.[1878]

1875 See, e.g., STANY Letter at 4; Nasdaq Letter IV at 2; Data Boiler Letter I at 81.

1876 The Commission estimates that approximately 98.5% of NMS stocks will have a round lot size of 100 shares. See supra Table 4.

1877 See supra Section V.B.2(a).

1878 See supra Section II.D.2(a).

(iii)Effects on Internalization of Retail Order Flow

The Commission believes that the change in the round lot size may have an effect on wholesalers in the retail order flow internalization business. Currently, some wholesalers,[1879] by arranging to execute orders on behalf of retail broker-dealers, offer superior prices relative to the existing NBBO (i.e., price improvement) to retail investors. As part of this arrangement, the wholesaler typically agrees that some percentage of the broker-dealer's orders will execute at prices better than the NBBO and/or agrees to certain execution quality metrics. The Commission expects that the new definition of a round lot will, at times, make the NBBO narrower for the affected stocks because the new definition will include orders that are at superior prices to the 100 share NBBO at a size less than 100 shares. As a result, it may become more difficult for the retail execution business of wholesalers to provide price improvement and other execution quality metrics at levels similar to those provided under the 100 share round lot definition today.[1880]

By the same mechanism, retail investors might or might not experience an improvement in execution quality, as measured by execution prices, from these wholesalers.[1881] Assuming that

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1879 See supra note 1797 for a discussion of wholesalers and retail internalization.

1880 Commenters agreed that a protected smaller round lot quote could affect the ability of internalizer to provide price improvement to retail investors. See, e.g., Virtu Letter at 5.

1881 This improvement may not be transparent to the retail investor. The price improvement metrics reported by retail broker-dealers do not take into account odd-lot quotes priced better than the NBBO. Even if a retail investor receives a better execution price from the new round lot definition, it might not show up as price improvement in retail wholesaler price improvement metrics if the NBBO also narrowed as a result of the new round lot size and now reflects odd-lot quotes that are priced better than the NBBO based on the current round lot size. One commenter stated retail wholesalers' price improvement metrics, along with Rule 605 statistics, are not accurate because they do not take into

the NBBO has narrowed and wholesalers continue to agree to provide the same amount of price improvement off of the narrower spread, retail investors will receive better execution prices. One commenter stated that retail investors will not receive better execution prices under the new round lot sizes because wholesalers already offer price improvement to retail investors that exceeds the potential improvements in the NBBO from the new round lot size.[1882] However, another commenter stated that all investors, including retail, would experience reduced execution costs from a tighter NBBO no matter where the execution took place.[1883] The Commission is uncertain whether the execution quality retail investors receive from wholesalers will change if the NBBO narrows for securities in the smaller round lot tiers because the effect of the amendments on retail execution quality would depend on how the change in the NBBO compared to the current price improvement offered by wholesalers, as well as on changes in the degree of price improvement wholesalers will offer in stocks with tighter NBBOs, which is uncertain.

To the extent that retail wholesalers are held to the same price improvement standards by retail broker-dealers in a narrower spread environment, the profitability of the retail execution business for wholesalers might decline. In particular, less "spread profit" would be available to the wholesaler in a narrower NBBO. This is, in part, because the wholesaler may often keep a

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account odd-lots quotes that are priced better than the NBBO. See Healthy Markets Letter I at 6-17.

1882 See TD Ameritrade Letter at 8, 10.

1883 See Best Ex Research Letter at 6 ("A tighter NBBO will reduce execution costs for all market participants—both retail and institutional investors—no matter where executions take place.").

portion of the spread profit that is not given as price improvement to the investor who submitted the order. Therefore, if the NBBO has narrowed and the same price improvement must still be provided, less revenue will be left for the wholesaler.[1884] To the extent this happens, it will be a transfer from the wholesaler to retail investors. As such, any impact on wholesaler profitability depends on the same factors as the impact on retail execution quality.

To make up for lower revenue per order filled in a narrower spread environment, wholesalers may respond by changing how they conduct their business in a way that may affect retail broker-dealers. There are several possibilities, including but not limited to, reducing per order costs associated with their internalization programs, such as reducing any payments for order flow or reducing the agreed upon metrics for price improvement. In the event that wholesalers reduce payments for order flow, retail broker-dealers may respond by changing certain aspects of their business. The Commission is uncertain as to how wholesalers may respond to the change in the round lot definition, and, in turn, how retail broker-dealers may respond to those changes, and the Commission is uncertain as to the extent of these effects.

The effect of lost revenue for wholesalers discussed above may be reduced if wholesalers currently use proprietary feeds to trade, to the extent they already see and respond to odd-lot quotations inside the NBBO and currently provide execution quality to customers based upon the superior odd-lot quotations.

[1884] The NBBO based off the new round-lot definition will be relevant to the spread considered by the wholesalers, because, among other things, it would be used for Rule 605 execution statistics. See infra Section V.C.1(b)(v) for further discussion of Rule 605 statistics.

(iv) Effects on Trading Venues

The Commission believes that changes in the NBBO caused by the new round lot definitions may also affect other trading venues, including exchanges and ATSs.[1885] Exchanges and ATSs have a number of order types that are based off of the national best bid and offer.[1886] Changes in the NBBO may affect how these order types perform and could also affect other orders they interact with. Some ATS matching engines also derive their execution prices based off of price improvement measured against the NBBO. Changes in the NBBO from the new round lot definition may affect execution prices on these platforms. Overall, the Commission believes that these interactions may affect relative order execution quality among different trading platforms, but it is uncertain of the magnitude of these effects.

Changes in relative execution quality may in turn affect the competitive standing among different trading venues, with trading venues that experience an improvement/decline in execution quality attracting/losing order flow. However, the Commission is uncertain of the magnitude of these effects.

One commenter stated that protecting the smaller round lot size could affect order flow to exchanges and other trading venues.[1887] The narrower protected NBBO in higher priced round lot tiers could cause more order flow to be routed to exchanges in these stocks. Because off-exchange trading venues would not be able to trade-through the NBBO, a narrower protected

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1885 See supra Section V.C.1(c)(iv) for additional discussion of effects on exchange rules.

1886 For example, the apparent price improvement over the NBBO calculated based off core data that is offered by a midpoint crossing network will be reduced as a result of changes to the NBBO.

1887 See Virtu Letter at 5.

NBBO would limit the price range in which off-exchange trading venues could execute trades and cause more orders to be routed to exchanges in order to not trade through a protected quote.

(v) Effects of Monthly Round Lot Calculation

The Commission believes that the use of the previous calendar month's average closing price on the primary listing exchange to determine the round lot tier for a given stock balances certain tradeoffs that should be considered when selecting such a benchmark.[1888] The Commission is balancing a more up-to-date stock price estimate against the costs imposed on market participants from having to frequently make updates to systems and practices to account for changes to a stock's round lot tier. A more recent average (e.g., the past week's average closing price) may better reflect the stock's current price level, and thereby lead to the stock being placed in the correct tier more frequently. However, such a recent estimate may be more volatile and thus more prone to causing frequent changes to the stock's status, especially if the stock's price level is close to a round lot tier cutoff point. This could impose a greater burden because it would require more frequent adjustments from market participants, including SROs and competing consolidators, to account for what a stock's round-lot tier is and what the NBBO for that stock would be given its tier.[1889]

Commenters stated that updating of a stock's round lot size each month could create confusion.[1890] One commenter stated that only updating a stock's round lot size monthly could

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1888 Commenters agreed that a monthly calculation strikes an appropriate balance. See MFA Letter at 10; Data Boiler I at 25.

1889 One commenter stated more frequent updates could impose a higher administrative burden. See NovaSparks Letter at 1.

1890 See, e.g., Nasdaq Letter IV at 17; MFA Letter at 12–13.

create confusion because it could lead to a stock's current price not reflecting its round lot tier, especially during months of extreme volatility or if a stock splits its shares.[1891] This commenter also stated that it could create confusion and uncertainty at the end of each month if a stock's price is close to a threshold and could also create confusion comparing Rule 605 statistics if a stock changed round lot tiers.[1892] The Commission does not believe that the updating a stock's round lot tier each month will create significant confusion. Most NMS stocks will still have a round lot size of 100 shares under the new round lot tier definitions.[1893] In response to comments, the Commission estimated that between August 2019 and August 2020, on average, only 17 stocks would change round lot tiers each month, which means that most stock's current prices would be reflective of their current round lot tiers. Additionally, primary listing exchanges will publish data on each stock's round lot size and the Commission expects market participants will modify or develop systems to automatically keep track of a stock's round lot size.[1894]

(vi) Costs of New Round Lot Definition

The Commission believes that the new round lot definition will impose two types of implementation costs on market participants: (1) one associated with upgrading systems to account for additional message traffic and (2) to modify and reprogram systems to account for the new round lot definition.

1891 See Nasdaq Letter IV at 17.

1892 See id.

1893 The Commission estimates that approximately 98.5% of NMS stocks will have a round lot size of 100 shares. See supra Table 4.

1894 See infra Section V.C.1(b)(vi) (discussing the implementation costs for these systems).

The Commission believes that market participants who currently rely solely on core data to obtain NBBO feeds will incur some infrastructure investment costs as a result of the change in the definition of a round lot. This is because the change will likely lead to more frequent updates to the NBBO and this will likely result in an increase in message traffic for NBBO feeds.[1895] Because most NMS stocks will still have a round lot size of 100 shares,[1896] the Commission does not believe the increase in message traffic will be significant. Therefore, the Commission does not believe that the system upgrades required by the new round lot definition will be significant. However, the Commission is unable to estimate the associated costs because it does not have access to information about the infrastructure expenses a broker-dealer incurs to process market data and because of the likelihood that such costs vary substantially according to the existing infrastructure of broker-dealers.

Additionally, for certain core data use cases, the costs described in the preceding paragraph are likely to be minimal. Many broker-dealers, when accessing data for the purposes of visual display, currently obtain NBBO quotes from the exclusive SIPs with a "per query" use case. This use case is set up so that a quote is only sent when it is asked for. The Commission believes that this setup has very little technological cost associated with it and that furthermore whatever cost there is to receiving such a feed will not be impacted by increasing the number of times the NBBO is updated over a given time period. Thus, the Commission believes that for

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1895 This will happen more in high-priced stocks where the new round lot definition will have more of an effect.

1896 The Commission estimates that approximately 98.5% of NMS securities will have a round lot size of 100 shares. See supra Table 4.

those broker-dealers who rely on per query use cases for their quotes, the upgrade costs resulting from changing the round lot definition will be minimal.[1897]

Trading venues and broker-dealers will experience implementation costs from having to modify and reprogram their systems, including matching engines and SORs, to account for the changes in the new round lot definition. Commenters stated that there would be implementation costs for market participants to develop systems to monitor and account for changes in a stock's round lot size.[1898] One commenter observed that broker-dealers would need to make changes to their order routing systems and systems that display customer orders each month to account for changes in the round lot size.[1899] This commenter also stated that regulators would need to modify their surveillance systems each month to account for changes in a stock's round lot size.[1900]

In the Proposing Release the Commission estimated that the implementation cost for a trading venue to update its systems, including its matching engine, to account for the new round lot definition and changes in the Order Protection Rule would be similar to the estimated costs of

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1897 This conclusion is contingent on the assumption that competing consolidators will choose to offer a per query service to market participants so that this arrangement may continue. Because a significant portion of market participants (particularly retail investors) access SIP data on a per query basis, the Commission believes that it is likely the Equity Market Data Plans will continue to charge fees on a per query basis and some competing consolidators will also offer a per query service in order to meet the demand of market participants.

1898 See, e.g., MEMX Letter at 4; Fidelity Letter at 6; STANY Letter at 4; Nasdaq Letter IV at 17; MFA Letter at 12–13; Angel Letter at 17.

1899 See MFA Letter at 12–13. This commenter stated that Rule 604 does not require a broker-dealer to display a customer's limit order if it is an odd-lot size.

1900 See id.